

The diagram illustrates the High-Level Decision-Making Layer, which is divided into two main sections: the Augmented Lagrangian-Based Discrete SAC and the Constrained Markov Decision Process.

Augmented Lagrangian-Based Discrete SAC

This section shows the flow of information from the State to the Action. The State is input to the State Space, which is then flattened and processed by an MLP. The output of the MLP is the Probability Distribution, which is then processed by the Original Action Space and the Valid Action Space to produce the Action. The Action is then fed back into the State Space. The diagram also shows the interaction between the Reward Critic, Actor, and Cost Critic, which are all part of the SAC framework.

Constrained Markov Decision Process

This section defines the components of the decision process. It includes the State Space and Action Space, which are defined by the Reward Function and Cost Function. The Reward Function is defined by Success, Comfort, Efficiency, and Safety. The Cost Function is defined by Risk Field and Hazardous Behavior. The diagram shows that the Action Space is constrained by the Cost Function, which is influenced by the Risk Field and Hazardous Behavior.

Low-Level Motion Planning Layer

Reference Waypoints

Model Predictive Control

Physical Constraints

Objective Function

Optimal Control Problem

Optimal U_p, X_p

If Action Unchecked

YES

NO

Motion Control

Acceleration a

Steering Angle δ

Action Shielding Mechanism

1 Safety Distance

2 Merging Collision

3 Merging Failure

If Action Unreplaced

YES

NO

Motion Control

Acceleration a

Steering Angle δ

Simulation Environments

Main Lane

Acceleration Lane